

SYSTEMS THINKING

Systems Analysis & Design

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Introduction to Systems Thinking
- 2 Systems Properties
- 3 Systems Classification



Outline

1 Introduction to Systems Thinking

2 Systems Properties

3 Systems Classification

2017

Hugging Face

Attention

Vector Database

$$w_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0.8$$

$$w_2 \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix} = 0.1$$

$$w_3 \begin{bmatrix} -0.3 \\ 0.9 \end{bmatrix} = 0.9$$

$$w_4 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Introduction to Systems Thinking I

- A **system** is a set of **interconnected elements** with a **common purpose**.

- Not all elements need to be connected to each other, but every **connection** should be meaningful.

- The more connections there are, the more complex the system becomes, but the representation must be feasible.

- Each element must have at least one connection. Isolated elements make no sense in a system.

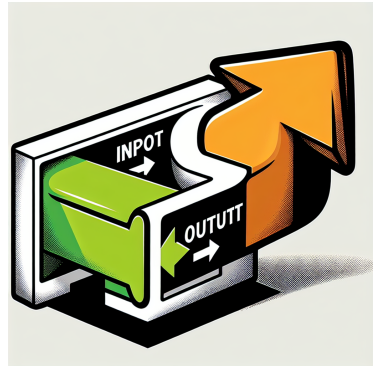
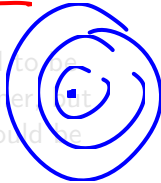


Figure: Prompt: Draw an image of a box with input and output arrows.



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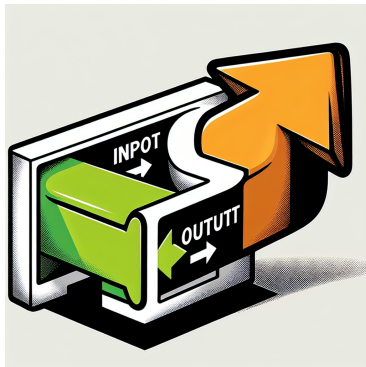


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clients

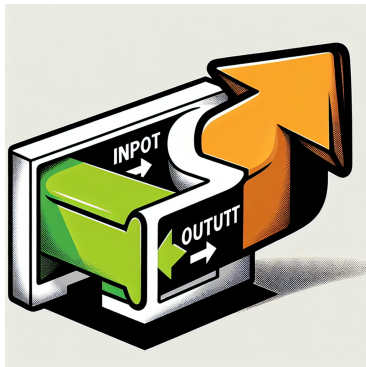


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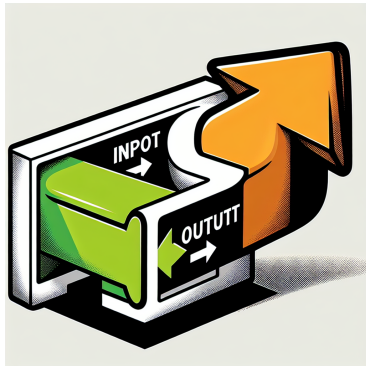
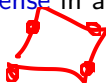


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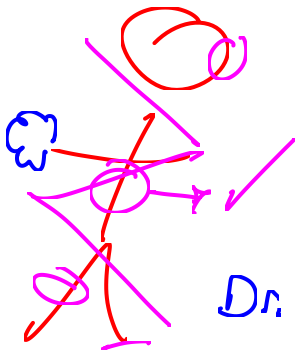


Complexity in Systems

System complexity could be defined as the number of elements and connections in a system.



Introduction to Systems Thinking II



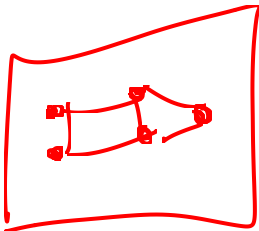
Dr. House

PHP → scalability?

- In **systems thinking**, if you just **split** parts and forget relationships, you will **lost the full picture**.
- It is called **holistic** approach, try to see **all the picture** with all the **meaning details**.
- Define the **box boundaries** is sometimes tricky, as we said, not too complex, not too simple. It is like the desired **universe balance of Thanos**.



Introduction to Systems Thinking II



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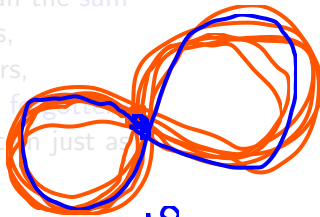
Introduction to Systems Thinking III

Header (The book)

- Another important concept is the homeostasis, it means to put a system in an equilibrium state. That is hard, systems are both not in equilibrium and resilient to change. Chaotic attractors study is useful here.

- A system is more than the sum of the parts. It means, relationships, behaviors, recovery capacity, are factors when you see the system just as a system.

Machine Learning

S-zo
Ins

Data

Dynamical

rule -

+ randomness



Introduction to Systems Thinking III

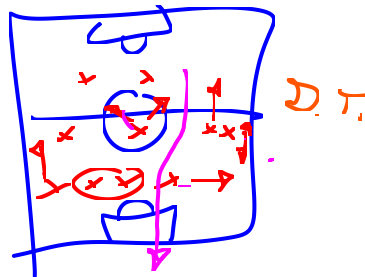
- Another important concept is the **homeostasis**, it means to put a system in an **equilibrium state**. That is hard, **systems** are both **not in equilibrium** and **resilient to change**. **Chaotic attractors** study is useful here.

- A **system** is more than the **sum of the parts**. It means, relationships, behaviors, recovery capacity, are **forgotten** when you see the system just as **its parts**.



Introduction to Systems Thinking IV

- **Systems thinking** is a way to understand and represent problems in order to find the best possible solution.
- Think in a **problem as a system** lets you understand details, involved elements, relevant information.
- Systems should be viable, auto-sostenible, provides internal feedback loops, and also looks like a whole live-entity.



Introduction to Systems Thinking IV

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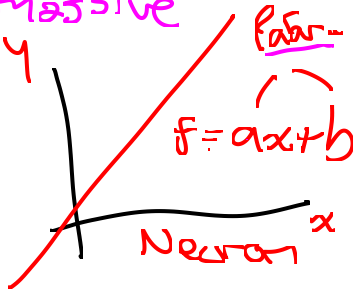
Introduction to Systems Thinking V

calculate mechanical/machine



massive

y



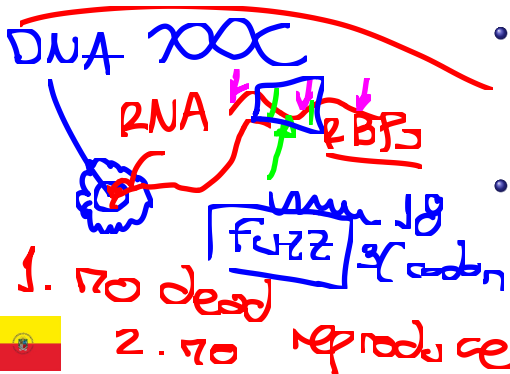
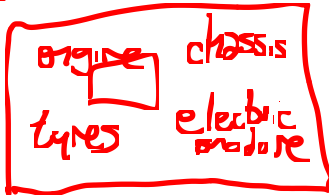
- **Computation** helps to represent behaviors in a mathematical way. Also, it lets to find patterns and information, simplify process; an example of all this is the Artificial Intelligence.

- Top-Down approach is useful when you want to see the full picture, and then split it into parts.
- Bottom-Up approach is useful when you want to see the parts and then connect them to get the full picture.



Introduction to Systems Thinking V

car

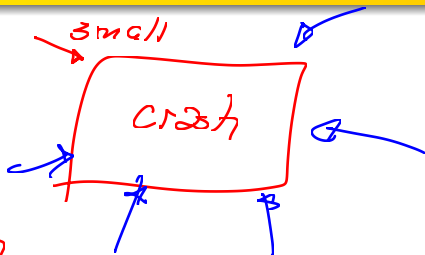


- **Computation** helps to **represent behaviors** in a mathematical way. Also, it lets to **find patterns** and information, simplify process; an example of all this is the **Artificial Intelligence**.
- **Top-Down** approach is useful when you want to **see the full picture**, and then split it into parts. 😊
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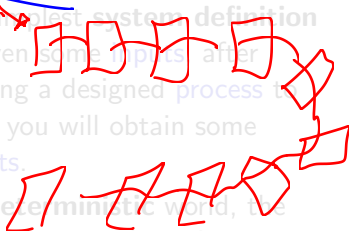


Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to making better decisions.



- The simplest system definition is: given some inputs, after applying a designed process to them, you will obtain some outputs.



- In a deterministic world, the same inputs produce the same outputs. However, real life is not deterministic.



Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to making better decisions.
- The simplest **system definition** is: given some **inputs**, after applying a designed **process** to them, you will obtain some **outputs**.
- In a **deterministic** world, the same **inputs** produce the same **outputs**. However, **real life is not deterministic**.

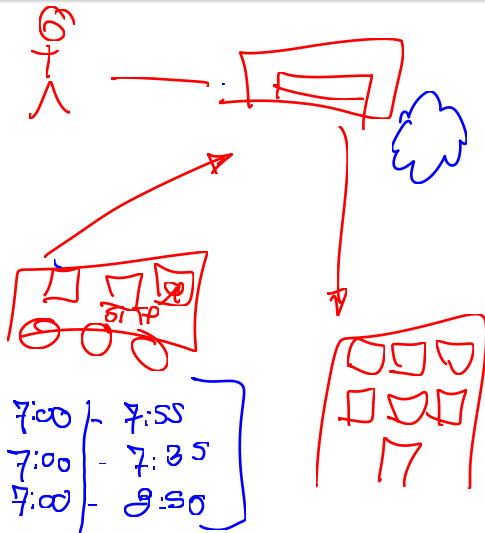


controlled conditions

$$F = m \cdot A.$$



Introduction to Systems Thinking VII

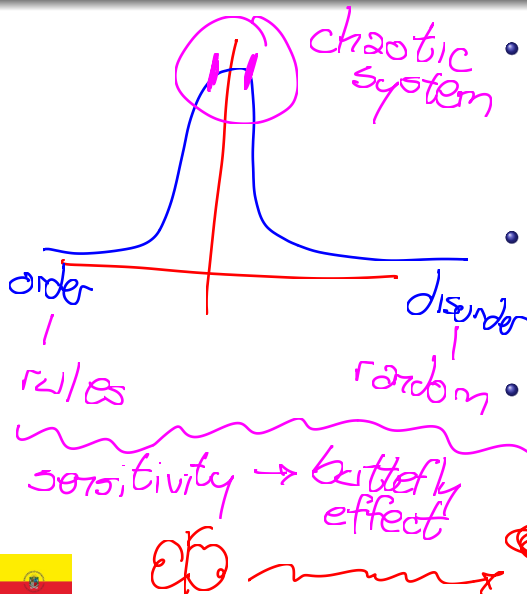


- Since **randomness** is normal in the real world, relying solely on deterministic processes is dangerous. Using **stochastic processes** is a better approach.
- Stochastic processes** make use of **probability**, which leads to a better representation of real-world behavior.

- Here **Chaos Theory** becomes a useful tool. To put it simply, chaos can be defined as a harmonious balance between rules and randomness.



Introduction to Systems Thinking VII

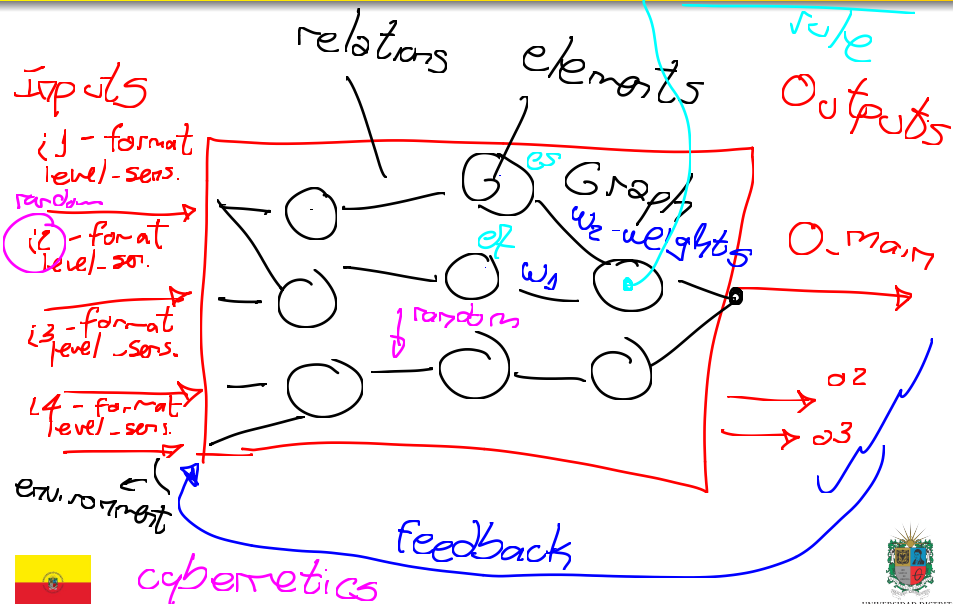


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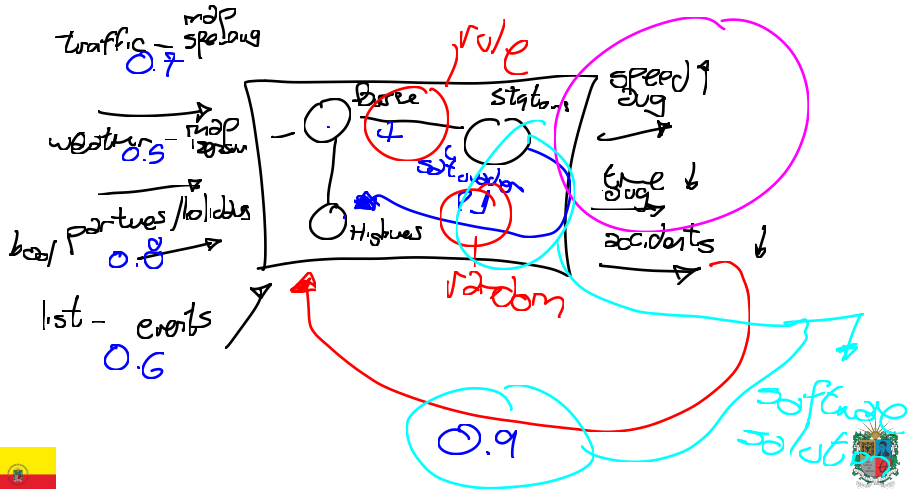


Systems Structure Draw

$e5, w2 + e7, w1$



Case of Study: Transportation System



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- 2 Systems Properties
- 3 Systems Classification

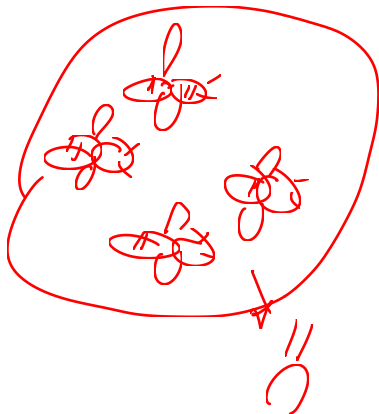


Systems Properties I

- **Emergence** is a property of systems that means that the whole system is more than the sum of its parts.

✓
synergy

- **Interconnectedness** is a property of systems that means that all the elements are connected in a meaningful way.
- **Feedback** is a property of systems that means that the system has internal loops that control the system behavior.
- **Hierarchy** is a property of systems that means that the system has levels of organization.



Systems Properties I

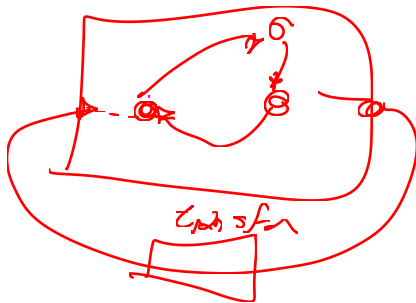
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isolated elements



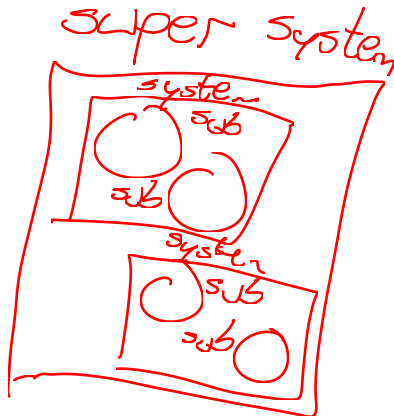
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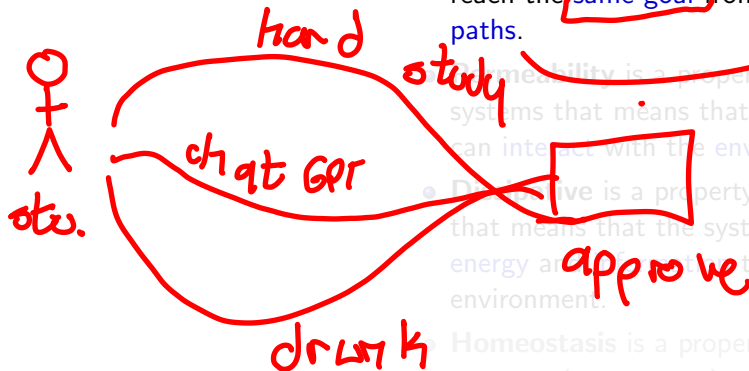
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Systems Properties II

- **Equifinality** is a property of systems that means that the system can reach the same goal from different paths.



- **Permeability** is a property of systems that means that the system can interact with the environment.
- **Dissipative** is a property of systems that means that the system can lose energy and information to the environment.

Homeostasis is a property of systems that means that the system can maintain a stable state.



Systems Properties II

Open
System

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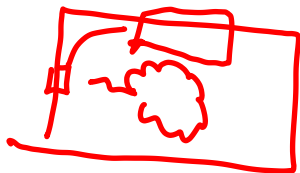
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Systems Properties II



Chaotic
Attractor

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Systems Properties III

- **Equilibrium** is a property of systems that means that the system can reach a stable state.
- **Adaptability** is a property of systems that means that the system can change to adapt to the environment.
- **Self-organization** is a property of systems that means that the system can organize itself.
- **Self-regulation** is a property of systems that means that the system can regulate itself.

Closed system

experiment



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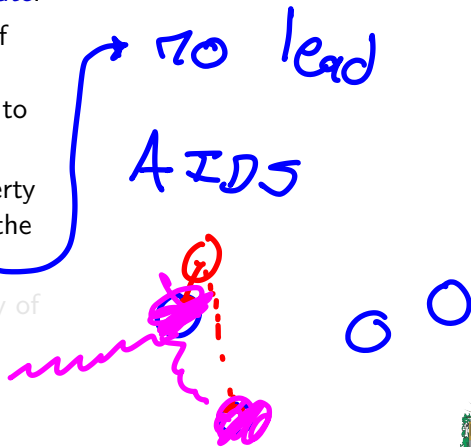
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Le Marc



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resources management



Outline

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Basic Concepts

A **system** can be classified according to different criteria like openness, adaptability, determinism, and linearity.



time



simulation



perme



Systems Classification I

- **Open systems** are systems that can **interact** with the **environment**.

→ nature → all

- **Closed systems** are systems that cannot interact with the environment.
- **Adaptive systems** are systems that can **change** to **adapt** to the environment.
- **Non-adaptive systems** are systems that cannot change to adapt to the environment.



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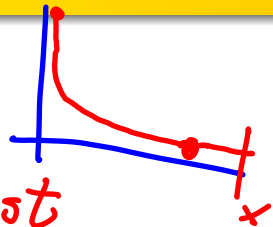
experiment
↓
equilibrium



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fast

small
iterations
amount



choice

3.7

oper

3.7

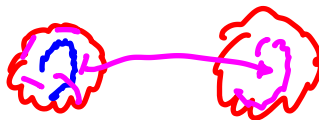
answer

adaptation



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→ Semi-adaptive



Systems Classification II

- **Deterministic systems** are systems that produce the same output for the same input.

Stochastic systems are systems that produce different outputs for the same input.

Linear systems are systems that produce proportional outputs for the same input.

- Non-linear systems are systems that produce non-proportional outputs for the same input.

Closed systems



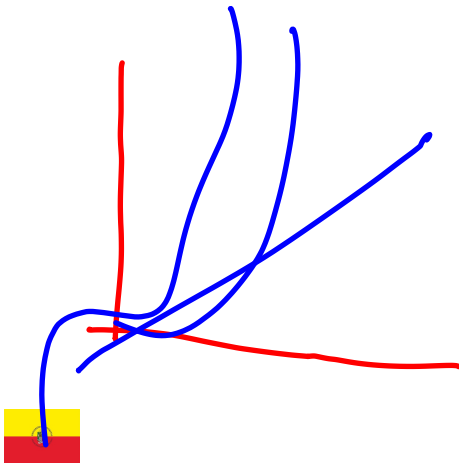
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random



Systems Classification II

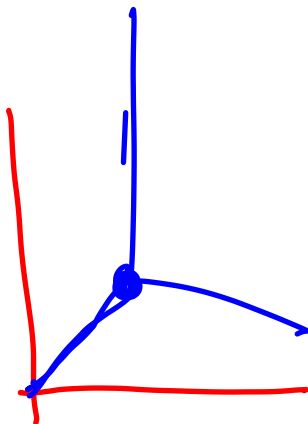


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$$f(x) = x$$



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non-linear
or
stochastic

Complex



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

